

Questions

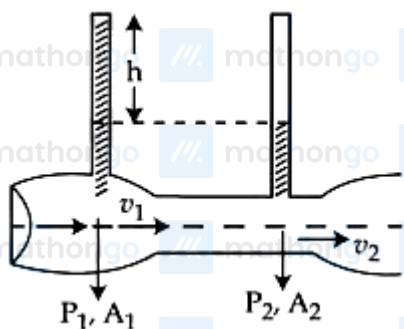
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Q1 - 2024 (04 Apr Shift 1)

Given below are two statements:

Statement I : When speed of liquid is zero everywhere, pressure difference at any two points depends on equation $P_1 - P_2 = \rho g (h_2 - h_1)$.

Statement II : In ventury tube shown $2gh = v_1^2 - v_2^2$



In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

Q2 - 2024 (04 Apr Shift 1)

A soap bubble is blown to a diameter of 7 cm. 36960 erg of work is done in blowing it further. If surface tension of soap solution is 40 dyne/cm then the new radius is _____ cm Take $\left(\pi = \frac{22}{7}\right)$

Q3 - 2024 (04 Apr Shift 2)

Given below are two statements :

Statement I : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

Statement II : The rise of a liquid in a capillary tube does not depend on the inner radius of the tube.

In the light of the above statements, choose the correct answer from the options given below:

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- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true.
- (3) Both Statement I and Statement II are false.
- (4) Both Statement I and Statement II are true.

Q4 - 2024 (04 Apr Shift 2)

Mercury is filled in a tube of radius 2 cm up to a height of 30 cm. The force exerted by mercury on the bottom of the tube is _____ N.

(Given, atmospheric pressure = 10^5 Nm^{-2} , density of mercury = $1.36 \times 10^4 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$, $\pi = \frac{22}{7}$)

Q5 - 2024 (05 Apr Shift 1)

Given below are two statements :

Statement I : When a capillary tube is dipped into a liquid, the liquid neither rises nor falls in the capillary. The contact angle may be 0° .

Statement II : The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

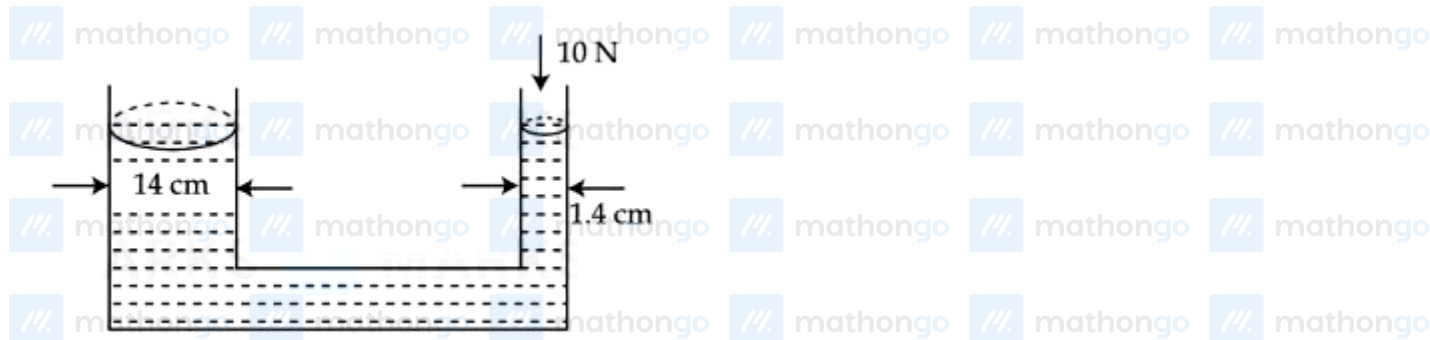
In the light of the above statement, choose the correct answer from the options given below.

- (1) Both Statement I and Statement II are false
- (2) Both Statement I and Statement II are true
- (3) Statement I is false but Statement II is true
- (4) Statement I is true and Statement II is false

Q6 - 2024 (05 Apr Shift 2)

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A hydraulic press containing water has two arms with diameters as mentioned in the figure. A force of 10 N is applied on the surface of water in the thinner arm. The force required to be applied on the surface of water in the thicker arm to maintain equilibrium of water is _____ N.

Q7 - 2024 (06 Apr Shift 1)

A small ball of mass m and density ρ is dropped in a viscous liquid of density ρ_0 . After sometime, the ball falls with constant velocity. The viscous force on the ball is :

- (1) $mg(1 - \rho\rho_0)$
- (2) $mg\left(1 + \frac{\rho}{\rho_0}\right)$
- (3) $mg\left(\frac{\rho_0}{\rho} - 1\right)$
- (4) $mg\left(1 - \frac{\rho_0}{\rho}\right)$

Q8 - 2024 (06 Apr Shift 1)

A big drop is formed by coalescing 1000 small droplets of water. The ratio of surface energy of 1000 droplets to that of energy of big drop is $\frac{10}{x}$. The value of x is _____

Q9 - 2024 (06 Apr Shift 2)

Pressure inside a soap bubble is greater than the pressure outside by an amount :

(given : R = Radius of bubble S = Surface tension of bubble)

- (1) $\frac{2S}{R}$
- (2) $\frac{4R}{S}$
- (3) $\frac{S}{R}$

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(4) $\frac{4S}{R}$

Q10 - 2024 (08 Apr Shift 1)

Correct Bernoulli's equation is (symbols have their usual meaning) :

(1) $P + mgh + \frac{1}{2}mv^2 = \text{constant}$

(2) $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

(3) $P + \rho gh + \rho v^2 = \text{constant}$

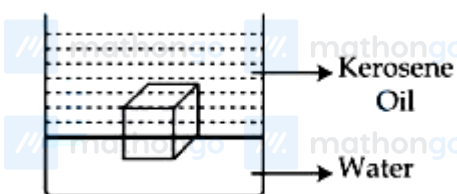
(4) $P + \frac{1}{2}\rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Q11 - 2024 (08 Apr Shift 1)

A liquid column of height 0.04 cm balances excess pressure of a soap bubble of certain radius. If density of liquid is $8 \times 10^3 \text{ kg m}^{-3}$ and surface tension of soap solution is 0.28 Nm^{-1} , then diameter of the soap bubble is _____ cm. (if $g = 10 \text{ m s}^{-2}$)

Q12 - 2024 (08 Apr Shift 2)

A cube of ice floats partly in water and partly in kerosene oil. The ratio of volume of ice immersed in water to that in kerosene oil (specific gravity of Kerosene oil = 0.8, specific gravity of ice = 0.9)



(1) 1 : 1

(2) 5 : 4

(3) 8 : 9

(4) 9 : 10

Q13 - 2024 (08 Apr Shift 2)

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Small water droplets of radius 0.01 mm are formed in the upper atmosphere and falling with a terminal velocity of 10 cm/s. Due to condensation, if 8 such droplets are coalesced and formed a larger drop, the new terminal velocity will be _____ cm/s.

Q14 - 2024 (09 Apr Shift 1)

A sphere of relative density σ and diameter D has concentric cavity of diameter d . The ratio of $\frac{D}{d}$, if it just floats on water in a tank is :

(1) $\left(\frac{\sigma-2}{\sigma+2}\right)^{1/3}$

(2) $\left(\frac{\sigma}{\sigma-1}\right)^{1/3}$

(3) $\left(\frac{\sigma-1}{\sigma}\right)^{1/3}$

(4) $\left(\frac{\sigma+1}{\sigma-1}\right)^{1/3}$

Q15 - 2024 (09 Apr Shift 2)

The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. The ratio between the volume of the first and the second bubble is:

(1) 1 : 9

(2) 1 : 3

(3) 1 : 27

(4) 1 : 81

Q16 - 2024 (09 Apr Shift 2)

A spherical ball of radius 1×10^{-4} m and density 10^5 kg/m³ falls freely under gravity through a distance h before entering a tank of water, If after entering in water the velocity of the ball does not change, then the value of h is approximately:

(The coefficient of viscosity of water is 9.8×10^{-6} N s/m²)

(1) 2296 m

(2) 2518 m

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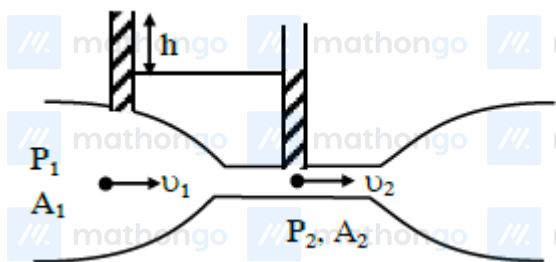
Answer Key

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Solutions

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Q1



Applying Bernoulli's equation

$$P_1 + \rho gh_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho gh_2 + \frac{1}{2}\rho v_2^2$$

$[h_1 \& h_2$ are height of point from any reference level]

Given $V_1 = V_2 = 0$ (for statement-1)

$$\therefore P_1 - P_2 = \rho g(h_2 - h_1)$$

For statement-2

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$P_1 - P_2 = \rho gh$$

$$P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$$

$$\rho gh = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$$

$$2gh = v_2^2 - v_1^2$$

Q2

$$\omega = \Delta U = S\Delta A$$

$$36960 \text{ erg} = \frac{40 \text{ dyne}}{\text{cm}} 8\pi \left[(r)^2 - \left(\frac{7}{2} \right)^2 \right] \text{ cm}^2$$

$$r = 7 \text{ cm}$$

Q3

Statement I is correct as we know contact angle depends on cohesive and adhesive forces.

Statement II is incorrect because height of liquid is given by $h = \frac{2T \cos \theta_c}{\rho g r}$ where r is radius of Tube

(assuming length of capillary is sufficient).

Q4

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Solutions

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$$\begin{aligned}
 F &= P_0 A + \rho_m g h A \\
 &= 10^5 \times \frac{22}{7} \times (2 \times 10^{-2})^2 \\
 &\quad + 1.36 \times 10^4 \times 10 \times (30 \times 10^{-2}) \left(\frac{22}{7} \times (2 \times 10^{-2})^2 \right) \\
 F &= 51.29 + 125.71 = 177 \text{ N}
 \end{aligned}$$

Q5

Capillary rise

$$h = \frac{2 T \cos \theta}{\rho g r};$$

If $\theta = 0^\circ$ then rise is non-zero $\therefore$ Statement-1 is incorrect.

Q6

$$\begin{aligned}
 \frac{F_1}{A_1} &= \frac{F_2}{A_2} \\
 \frac{F_1}{\pi(7)^2} &= \frac{10}{\pi \times (0.7)^2} \\
 F_1 &= 1000 \text{ N}
 \end{aligned}$$

Q7

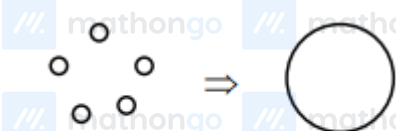
$$mg - F_B - F_v = ma$$

 $a = 0$ for constant velocity

$$mg - F_B = F_v$$

$$F_v = mg - v \rho_0 g = mg - \frac{m}{\rho} \rho_0 g = mg \left(1 - \frac{\rho_0}{\rho} \right)$$

Q8



1000 drops

Big drop

$$1000 \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3$$

$$10r = R$$

$$R = 10r$$

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Solutions

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$$\frac{\text{S.E. of 1000 drops}}{\text{S.E. of Big drop}} = \frac{1000 (4\pi r^2) T}{4\pi R^2 T}$$

$$= \frac{1000 \times r^2}{(10r)^2} = 10 = \frac{10}{x}$$

$$\therefore x = 1$$

Q9

There are two liquid-air surfaces in bubble so

$$\Delta P = 2 \left(\frac{2S}{R} \right) = \frac{4S}{R}$$

Q10

$$P + \rho gh + \frac{1}{2} \rho V^2 = \text{constant}$$

Q11

$$\rho gh = \frac{4S}{R}$$

$$\Rightarrow R = \frac{4 \times 0.28}{8 \times 10^3 \times 10 \times 4 \times 10^{-4}}$$

$$\Rightarrow \frac{0.28}{8} \text{ m} = \frac{28}{8} \text{ cm}$$

$$\Rightarrow R = 3.5 \text{ cm}$$

$$\text{Diameter} = 7 \text{ cm}$$

Q12

 v_1 = volume immersed in water. v_2 = volume immersed in oil.

$$v_1 \rho_w g + v_2 \rho_o g = (v_1 + v_2) \rho_c g$$

$$v_1 + \frac{v_2 \rho_o}{\rho_w} = (v_1 + v_2) \frac{\rho_c}{\rho_w}$$

$$= v_1 + 0.8v_2 = 0.9v_1 + 0.9v_2$$

$$= 0.1v_1 = 0.1v_2$$

$$v_1 : v_2 = 1 : 1$$

Q13

 m = mass of small drop M = mass of bigger drop

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Solutions

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$$V_t = \frac{2}{9} \frac{R^2(\rho - \sigma)g}{\eta}$$

$$8 \propto m = M$$

$$8r^3 = R^3 \Rightarrow R = 2R$$

as $V_t \propto R^2 \therefore$ Radius double so V_t becomes 4 time

$$\therefore 4 \times 10 = 40 \text{ cm/s}$$

Q14

$$\text{weight } (w) = \frac{4}{3}\pi \left(\frac{D^3 - d^3}{8} \right) \sigma g$$

$$\text{Buoyant force } (F_b) = 1 \times \frac{4}{3}\pi \left(\frac{D^3}{8} \right) \cdot g$$

$$\text{For Just Float} \Rightarrow w = F_b$$

$$\Rightarrow (D^3 - d^3) \sigma = D^3$$

$$\Rightarrow 1 - \frac{d^3}{D^3} = \frac{1}{\sigma}$$

$$\Rightarrow 1 - \frac{1}{\sigma} = \left(\frac{d}{D} \right)^3$$

$$\Rightarrow \left(\frac{\sigma}{\sigma - 1} \right)^{\frac{1}{3}} = \left(\frac{D}{d} \right)$$

Q15



$$P_1 - P_0 = \frac{4T}{r_1}$$

$$P_2 - P_0 = \frac{4T}{r_2}$$

$$P_1 - P_0 = 3(P_2 - P_0)$$

$$\frac{4T}{r_1} = 3 \frac{4T}{r_2}$$

$$r_2 = 3r_1$$

$$\frac{V_1}{V_2} = \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{1}{27}$$

Q16

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Solutions

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$$V_T = \frac{2gR^2[\rho_B - \rho_L]}{9\eta}$$

$$\Rightarrow V_T = \frac{2}{9} \times \frac{10 \times (10^{-4})^2}{9.8 \times 10^{-6}} [10^5 - 10^3]$$

$$\Rightarrow V_T = 224.5$$

when ball fall from height (h)

$$V = \sqrt{2gh}$$

$$h = \left(\frac{V^2}{2g} \right) = 2518 \text{ m}$$

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